

Article

Causation Analysis of Marine Traffic Accidents Using Deep Learning Approaches: A Case Study from China's Coasts

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Abstract: In response to the increasing frequency of maritime traffic accidents along China's coast, this study develops an accident-cause analysis framework that integrates an optimized Bidirectional Encoder Representations from Transformers (BERT) with a Bidirectional Long Short-Term Memory network (BiLSTM), combined with the Apriori association rule algorithm. Systematic performance comparisons demonstrate that the BERT + BiLSTM architecture achieves superior unstructured-text-processing capability, attaining 89.8% accuracy in accident-cause classification. The hybrid framework enables comprehensive investigation of complex interactions among human factors, vessel characteristics, environmental conditions, and management practices through multidimensional analysis of accident reports. Our findings identify improper operations, fatigue-related issues, illegal modifications, and inadequate management practices as primary high-risk factors while revealing that multi-factor interaction patterns significantly influence accident severity. Compared with traditional single-factor analysis methods, the proposed framework shows marked improvements in Natural Language Processing (NLP) efficiency, classification precision, and systematic interpretation of cross-factor correlations. This integrated approach provides maritime authorities with scientific evidence to develop targeted accident prevention strategies and optimize safety management systems, thereby enhancing maritime safety governance along China's coastline.

Keywords: causation analysis; marine traffic accident; BERT + BiLSTM; Apriori association rule; deep learning model



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1. Introduction

Shipping stands as one of the most international of the world's major industries, yet it remains inherently perilous [1]. According to the Statistical Bulletin on the Development of the Transportation Industry, between 2013 and 2023, China's coastline saw 1898 major accidents, leading to 2034 fatalities or disappearances and the sinking of 849 vessels [2]. On a global scale, the situation is similarly concerning; from 2012 to 2021, Allianz Global Corporate & Specialty (AGCS) managed an average of approximately 2700 maritime accident insurance cases per year¹. Given the phenomenon of underreporting and concealment of accidents, the actual accident number should be higher than the published data. Due to its unique environmental characteristics, marine traffic gives rise to a diverse array of accident

types, including ship collisions, self-sinkings, fire and explosions, oil spill pollution, and others [3]. These incidents may not only result in direct casualties and property losses of vessels but also trigger cascading environmental crises [4]. Additionally, the growing demand for water transportation has resulted in a more complex navigation environment, thereby increasing the safety risks associated with waterborne navigation [5]. Consequently, it is critical to conduct a thorough analysis of the causes of maritime traffic accidents to reduce accident rates and minimize related losses.

This paper is organized as follows: Section 2 provides a comprehensive literature review and highlights the contribution. Section 3 introduces the accident classification model and the ship cause analysis method. Section 4 trains the model based on relevant data and uses Chinese marine traffic accidents as actual cases for research. Section 5 analyzes and discusses the causes of accidents. Section 6 concludes the major findings and provides prospects for further research.

2. Literature Review

2.1. Investigation into Causative Factors

For an extended period, accident-cause analysis has remained a central focus in safety research. Numerous scholars have examined maritime traffic accident causes from diverse perspectives, with the aim of identifying effective risk control strategies. These studies attribute accident causation to four primary factors—human, vessel, environment, and management—with each factor’s importance varying across different contexts [6]. Among them, human error, pointed out by the UK Marine Accident Investigation Branch (MAIB), is the main cause of most water accidents [7]. Subsequent studies have demonstrated that over 80 percent of maritime accidents can be directly attributed to human error or result from crew members’ inappropriate responses to hazardous situations [8]. For example, Chowdhury et al. found that human error in engine room operations was one of the important causes of accidents [9]. Hu et al. integrated the Human Factors Analysis and Classification System (HFACS) framework with Structural Equation Modeling (SEM) to quantitatively assess the layered interactions among human factors—particularly resource management—in relation to the causality of marine traffic accidents [10]. Given the key position of human factors, in-depth research on them is extremely necessary.

However, the causes of maritime accidents are multifaceted, so considering only human factors is insufficient. Many studies have moved beyond the singular focus on human errors. For instance, Bhardwaj et al. identified that management problems often underlie the root causes of accidents [8]. Kokotos et al. demonstrated that the implementation of an effective assessment system significantly influences the occurrence of ship accidents [11]. Cao et al. established a clear causal relationship between management practices and accident severity [12]. In addition to these human and managerial dimensions, objective factors such as ship-related issues and environmental conditions warrant further exploration. The International Maritime Organization has reported that approximately one-quarter of maritime accidents are attributable to ship mechanical failures [13]. Kimera et al. discovered that ship factors are closely linked to accident severity [14], a relationship that is particularly pronounced in collision incidents [15,16]. Regarding environmental factors, Chen et al. argues that good navigational conditions help to reduce the probability of serious accidents [17]. Lin et al. observed that certain regions exhibit a heightened risk of collision [18], while Jon et al. underscored the significant influence of environmental conditions on accident severity [19]. Bye directly states that low environmental visibility increases the likelihood of accidents [20].

In summary, the existing literature on maritime accident causation has predominantly focused on human factors as the primary contributors to incidents. However, these studies

tend to examine factors in isolation, offering limited insight into the complex interaction mechanisms among multiple causal elements in maritime accidents.

2.2. Investigation into Causative Methodologies

During the systematic exploration of maritime accident causes, scholars have persistently endeavored to develop more efficient and precise analytical methodologies. The Formal Safety Assessment (FSA) method, officially endorsed by the International Maritime Organization, has become a widely adopted approach in risk management and accident analysis [21]. Bayesian networks (BNs) have demonstrated unique value in maritime safety studies due to their robust capability to represent complex relationships [22]. For example, Khan et al. employed Bayesian network technology to evaluate the risks associated with the transportation of dangerous goods in port environments [23]. Scholars were also dedicated to enhancing the Bayesian formula to address more complex research requirements [24]. However, given their reliance on substantial prior probability information, these methods often need to be integrated with other approaches to effectively mitigate subjective bias [25].

Furthermore, scholars have explored accident causation by developing a variety of algorithms and models. For instance, Chin et al. employed an ordered probity regression model to predict the risk of ship collisions [26]. Feng et al. proposed a quantitative method that integrates information entropy with K-means clustering to evaluate ship collision risk areas [27]. Melnyk et al. developed a causal model, in conjunction with cluster analysis, to systematically identify critical risk factors for maritime accidents from a data-driven perspective [28]. Based on the Fuzzy SWOT-AHP approach, Kecici et al. devised a cause analysis technology for ship accidents, termed Ship Accident Root cause Evaluation (SHARE) [29]. Chen et al. utilized the Reason-SHEL model to examine human factors in ship accidents [30]. Ma et al. constructed a dynamic modeling system for analyzing human factors in maritime collisions by integrating the Decision-Making Trial and Evaluation Laboratory (DEMATEL) model with fuzzy cognitive mapping, grounded in the HFACS framework [31].

With advancements in computer technology, NLP has emerged as an increasingly vital tool in accident analysis. Its application spans several domains, including public opinion monitoring [32], aviation [33], and mining [34]. Deep learning, with its exceptional data processing capabilities, has significantly advanced natural language processing and attracted considerable attention within the maritime domain. For example, Liu et al. employed an optimized Recurrent Neural Network (RNN) to forecast regional collision risks over a short time horizon in a specific ocean area [35]. Gan et al. introduced a cause-and-effect prediction model for marine accidents based on convolutional networks utilizing mutual information depth maps, achieving high-precision prediction and ranking of accident causes, thereby offering valuable insights for risk assessment and prevention [36]. Knowledge graphs have also garnered interdisciplinary interest. Chen et al. combined deep learning with the Classification and Regression Trees algorithm to develop an accident knowledge graph and identify critical risk factors through triplet extraction [37]. Gan et al. applied NLP to extract textual information from accident reports and construct knowledge graphs [38]. NLP is extensively utilized for data extraction tasks across various domains. For instance, Nurduhan et al. employed BERT-based quantization, UMAP reduction, and K-Means clustering to analyze human factors in maritime accidents, thereby visualizing their interactions and causal relationships [39]. Huang et al. utilized knowledge extraction to identify accident-related entities and elucidate the logical pathways of waterway accidents involving dangerous goods through clustering [40]. Yan et al. analyzed the semantic relationships between hazards in maritime accidents and developed a semi-automatic

model for hazard and accident cause identification grounded in BERT [41], although this study placed less emphasis on the correlation between factors.

To sum up, the intricate interplay among multiple factors remains insufficiently elucidated, indicating a need for more comprehensive and systematic approaches in accident analysis. Future research should integrate data-mining techniques with deep learning methodologies to extract latent information from accident reports and reveal complex interdependencies, thereby enhancing accident prevention and management strategies. Moreover, a detailed synthesis of the reviewed literature is provided in Table 1.

Table 1. Literature summary.

Research Classification	Object of Primary Concern	Methodology	Research Focus
Subjective Factor	Human factor, Management factor	Case Study, Quantitative Analysis of HFACS, Structural Equation Modeling, Causality Analysis, and Statistical Analysis.	The critical role of human error and management deficiencies in accidents is analyzed, and the interplay among multiple factors is elucidated.
Objective Factor	Ship factor, Environmental factor	Statistical Data Analysis, Correlation Analysis, Empirical Analysis, Case Study Analysis, and Comprehensive Statistical Evaluation.	Analyze the influence of mechanical failures and environmental factors, such as low visibility, on the occurrence and severity of accidents.
Traditional Analysis Methods	Comprehensive Analysis of Accident Risk	Formal Security Assessment Method, Regression Model, Causal Model Combined with Cluster Analysis.	Assess accident risks and propose a conventional methodology for systemic risk management and accident prevention.
Data-driven Models	Multifactor Comprehensive Analysis	Bayesian Networks (BNs) and Their Optimization Methodologies, Fuzzy SWOT AHP, Reason-SHEL, DEMATEL.	Characterize and assess the complex interrelationships among factors involved in accidents to minimize subjective bias. Conduct dynamic analyses of human factors and causal mechanisms in accidents to systematically explore multi-factor interactions.
Deep Learning	Accident Text Information Extraction	Feature Extraction and Knowledge Mapping.	Improve the accuracy of accident risk prediction, achieve the extraction of information from accident report texts, and enable efficient identification of causal relationships.

This study distinguishes itself by first addressing the limitations of traditional single-factor analyses, which tend to focus exclusively on human error or equipment failure, thereby overlooking the complex interplay among various contributing factors. Recognizing the multifaceted nature of maritime accidents, our approach integrates a novel analytical framework that leverages advanced deep learning and data-mining techniques. Specifically, the framework combines a BERT + BiLSTM deep learning model with the Apriori association rule algorithm to automatically extract deep semantic information from unstructured accident reports. This integration not only enhances classification accuracy and interpretability but also reveals the intricate interdependencies among multiple risk factors.

Building on this innovative model, this study employs a two-stage process. Initially, accidents are categorized based on the intrinsic characteristics of their associated texts—a strategy that enhances classification precision and facilitates the effective analysis of large datasets. An extensive, internationally sourced dataset is utilized to train the model, which is subsequently refined and applied to Chinese maritime accident reports. This sequential methodology establishes a systematic and robust analytical framework, directly addressing the recurrent incidence of marine traffic accidents and their severe impacts on personnel, property, and environmental resources. In summary, this work lays a solid scientific foundation for maritime safety management and accident prevention strategies, offering valuable methodological insights for the application of natural language processing and data-mining techniques in complex accident analysis. The proposed framework holds significant relevance for maritime regulatory authorities, research institutions, scholars, and safety risk analysis professionals.

3. Methodology

This study focuses on the maritime sector, where accident reports are rich in technical terminology—such as sea state descriptions, ship status, and navigation operations—those traditional methods often overlook. BERT adapts well to this specialized vocabulary, enabling direct fine-tuning with maritime data without the need to train deep networks from scratch. This not only conserves computational resources but also rapidly produces high-quality text vector representations. In the maritime context, accurately capturing the sequence of events and their causal relationships is critical. BiLSTM excels at extracting bidirectional sequence information, making it particularly effective for tasks that demand sensitivity to temporal and causal dynamics. Moreover, because marine accidents typically result from multiple interacting factors, the Apriori algorithm is employed to efficiently identify frequent co-occurrence patterns within extensive datasets. Its flexibility in adjusting support and confidence thresholds facilitates the discovery of underlying causal relationships.

By integrating these approaches, the framework leverages BERT's robust semantic capabilities alongside BiLSTM's sequence modeling and Apriori's association rule mining, offering a novel perspective on elucidating the complex multi-factor interactions that underlie maritime accidents.

3.1. BERT + BiLSTM Classification Model

The BERT + BiLSTM model synergistically combines BERT's powerful semantic encoding capabilities with BiLSTM's proficiency in capturing sequential information. In this framework, BERT initially encodes the accident-related text into high-dimensional contextual vectors, effectively extracting deep semantic features. Subsequently, BiLSTM processes these vectors to capture sequence dependencies and contextual associations, after which a linear classifier completes the accident-cause classification. This dual-dimensional analysis—integrating both semantic richness and sequential dynamics—significantly enhances the model's classification accuracy and generalization ability, thereby improving overall performance.

3.1.1. BERT

BERT (Bidirectional Encoder Representations from Transformers) is based on the Transformer architecture [42] and is a pre-trained model for NLP proposed by Google [43].

(1) Input Layer

The tokenizer will embed [CLS] tokens at the beginning of the sequence to indicate that this is a classification task and embed tokens at the end of the sequence or sentence to indicate the end of the sequence. Thus, a new sequence S' is constructed:

$$S' = [[CLS], S_1, S_2, \dots, S_n, [SEP]] \quad (1)$$

Among them, $S_1, S_2, \dots, S_n$ are the tokens in the original token sequence S .

(2) Embedding layer

For the input token sequence $S = [S_1, S_2, \dots, S_n]$, the model first maps it to a high-dimensional vector representation $E(S_i)$ with the help of the embedding matrix E and then adds position embedding to give position information to each token to help the model capture the order relationship. The output of the embedding layer is the sum of the word embedding and position embedding of each token, forming the initial representation of the sequence.

(3) Transformer Encoder Layer

Transformer uses a multi-head attention mechanism to increase the model's ability to focus on different positions [44]. The query, key, and value matrices are split into multiple heads, and the self-attention operations are performed separately and then the outputs are concatenated:

$$Attention(Q, K, V) = softmax\left(\frac{QK^T}{\sqrt{d_k}}\right)V \quad (2)$$

$$MultiHead(Q, K, V) = Concat(head_1, \dots, head_h)W^O \quad (3)$$

$$FFN(H) = max(0, HW_1 + b_1)W_2 + b_2 \quad (4)$$

$$Output_{Sublayer} = LayerNorm(H + Sublayer(H)) \quad (5)$$

where each head $head_i = Attention(QW_i^Q, KW_i^K, VW_i^V)$ and W^O is the parameter matrix of the output linear transformation. W_1, W_2 are the weight matrix, and b_1, b_2 are the bias vectors. $Sublayer(H)$ is a self-attention layer or a feedforward neural network.

3.1.2. BiLSTM

In a recurrent neural network, a long short-term memory network (LSTM) cell receives the hidden state h_{t-1} and cell state C_{t-1} of the previous time step and the input x_t of the current time step at each time step. It outputs the hidden state h_t and cell state C_t of the current time step.

Key components of LSTM include the following [45]:

$$i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i) \quad (6)$$

$$f_t = \left(W_f \cdot [h_{t-1}, x_t] + b_f\right) \quad (7)$$

$$o_t = \sigma(W_o \cdot [h_{t-1}, x_t] + b_o) \quad (8)$$

$$\tilde{C}_t = \tanh(W_C \cdot [h_{t-1}, x_t] + b_C) \quad (9)$$

$$C_t = f_t * C_{t-1} + i_t * \tilde{C}_t \quad (10)$$

$$h_t = o_t * \tanh(C_t) \quad (11)$$

Among them, σ is the sigmoid activation function, W_i is the weight matrix, b_i is the bias, W_f is the weight matrix, b_f is the bias, $\tanh$ is the hyperbolic tangent activation function, W_C is the weight matrix, and b_C is the bias. BiLSTM is composed of a forward

LSTM and a reverse LSTM. The forward LSTM is responsible for input, and the reverse LSTM is responsible for extracting features in forward and reverse order. The feature vectors of the two outputs are then combined into the final feature expression.

For instance, consider the sentence: “Thick fog reduces visibility at sea, making it challenging for crew members to distinguish navigation aids and lights”. In the input layer, BERT tokenizes this sentence using Word Piece, yielding the sequence: [“[CLS]”, “thick”, “fog”, “reduces”, “visibility”, “at”, “sea”, “,”, “making”, “it”, “challenging”, “for”, “crew”, “members”, “to”, “distinguish”, “navigation”, “aids”, “and”, “lights”, “[SEP]”]. In the embedding layer, each token is mapped to a high-dimensional vector that combines three components: word embeddings (capturing vocabulary semantics, e.g., “fog” relates to weather), position embeddings (indicating the token’s order in the sequence, e.g., “visibility” is the fifth token), and segment embeddings (marking the sentence segment; all tokens belong to segment A for a single sentence). The resulting matrix is then processed by the Transformer layer, where each token initially has a context-free representation. Through a multi-layer self-attention mechanism, the Transformer progressively extracts contextual features; for example, “visibility” is associated with “thick fog” and “sea” to reflect its context, while “challenging” connects with “crew members” and “distinguish” to capture the complexity of the task. A subsequent feedforward network further refines these nonlinear characteristics, updating the vectors to include global contextual information (e.g., recognizing “navigation aids” and “lights” as critical navigational elements).

Afterward, the BiLSTM layer processes these vectors bidirectionally. The forward LSTM captures left-to-right dependencies (such as the influence of “crew members” on the subsequent action “distinguish”), while the backward LSTM captures right-to-left dependencies (such as “lights” being modified by “navigation aids”). This integration of bidirectional temporal features reinforces the causal relationships—for instance, between “reduces visibility” and “challenging”. Finally, the output layer applies a Softmax function to compute the probability distribution, yielding the final classification decision (in this case, “poor visibility”). The complete structure of the model is shown in Figure 1.

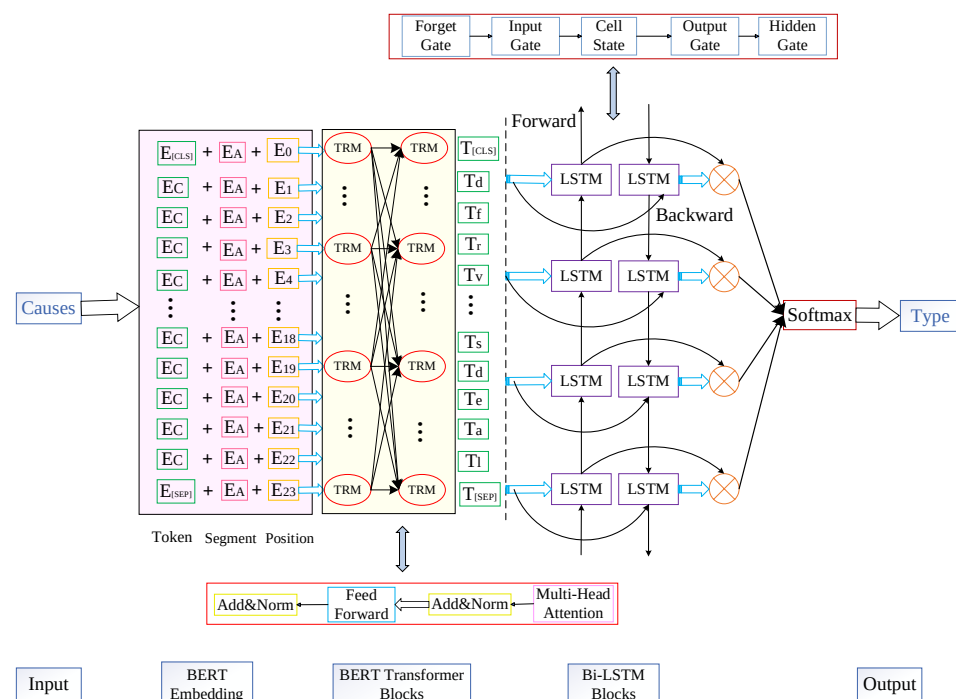


Figure 1. Model structure.

3.2. Apriori Algorithm

The Apriori association rule algorithm is a classic algorithm for discovering frequent item sets and association rules in accident databases [46]. This framework can explore potential correlations between multiple factors in accident data, uncover the interactive relationships behind accident causes, address the limitations of classification models in causal analysis, and improve the overall understanding of the factors contributing to accidents [47]. The main process is to generate frequent item sets and then generate association rules as follows:

$$\text{Support}(X \cup Y) = \frac{\text{Number of transactions in the frequent itemset}(X \cup Y)}{\text{Total number of transactions}} \quad (12)$$

$$\text{Confidence}(X \Rightarrow Y) = \frac{\text{Support}(X \cup Y)}{\text{Support}(X)} \quad (13)$$

$$\text{Lift}(X \Rightarrow Y) = \frac{\text{Confidence}(X \Rightarrow Y)}{\text{Support}(Y)} \quad (14)$$

$$\text{Conviction}(X \Rightarrow Y) = \frac{1 - \text{Support}(Y)}{1 - \text{Confidence}(X \Rightarrow Y)} \quad (15)$$

The set of maximum frequent items is found according to the graph and the association rules are composed based on the threshold restrictions (see Figure 2).

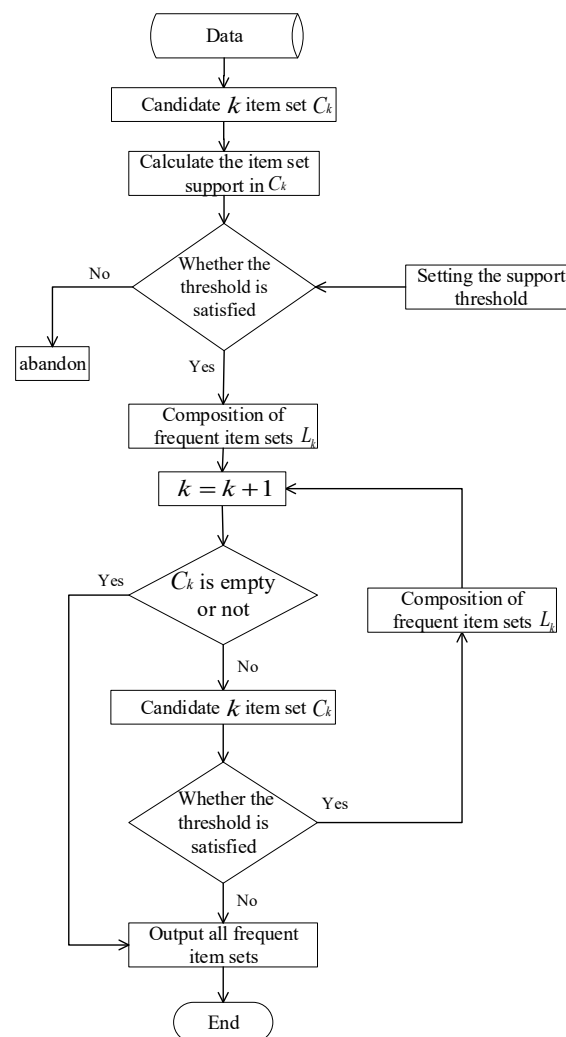


Figure 2. Association rules flowchart.

3.3. Framework

Although the BERT + BiLSTM model provides deep semantic understanding and efficient classification of accident-related texts, it does not account for the interactions among contributing factors. In contrast, the Apriori algorithm excels at revealing complex multi-factor interactions by mining association rules. By integrating these two approaches, the model delivers high-quality input data for rule mining, while the association algorithm supplements the classification output with insights into inter-factor relationships. This synergy not only enhances classification accuracy but also uncovers hidden multi-factor associations, thereby improving the comprehensiveness and practical applicability of accident analysis. Consequently, this study proposes a maritime accident analysis framework that combines an enhanced deep learning BERT model with Apriori association rules (see Figure 3).

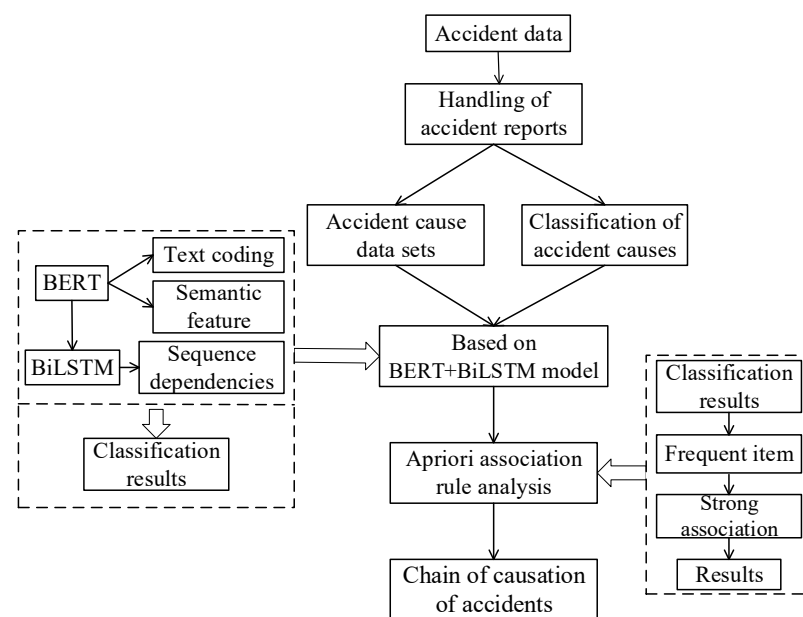


Figure 3. Research framework chart.

4. Case Study

4.1. Data

The accident report data utilized in this study were primarily sourced from the official website of the China Maritime Safety Administration, reports issued by local maritime safety bureaus (covering the period from 2014 to 2024), and the official website of the Global Integrated Shipping Information System (GISIS). Notably, GISIS data were predominantly employed for model training; therefore, greater emphasis was placed on the content and quality of the data rather than the specific temporal range. The core analysis relied on data reports from China's coastal regions. Furthermore, critical information within the accident reports was extracted using regular expressions. For instance, a report might describe "Accident location: longitude 121.50° E, latitude 29.80° N; Accident type: Collision". Regular expressions were applied to separately extract longitude, latitude information, and accident types to ensure precise data extraction. During data processing, it was discovered that some reports contained missing information. For such cases with incomplete data, an initial manual review of the original reports was conducted to supplement the missing details. If essential information remained unavailable even after consulting the original reports, the corresponding data were excluded to maintain overall data integrity and quality. The specific steps are represented in Figure 4.

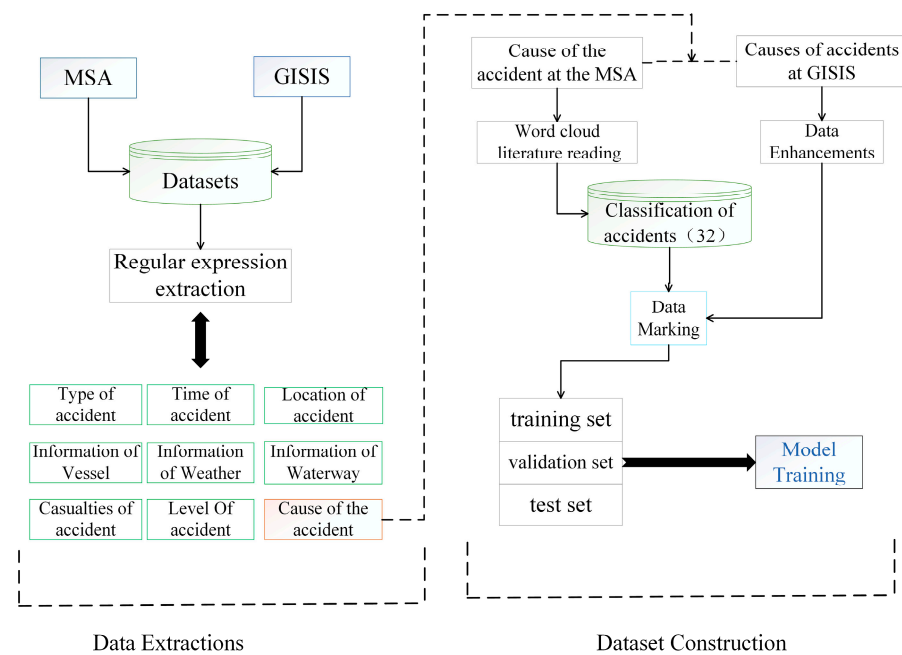


Figure 4. Data processing.

In terms of classification, given that accident causes typically result from multiple factors, this study initially employed word cloud analysis to visually represent the frequency of causes mentioned in Chinese accident reports. While this method provides an intuitive overview of prevalent causes, it lacks detailed classification dimensions. Consequently, based on relevant studies [6,41,48–50], the causes were systematically categorized into four primary groups—human factors, ship factors, environmental factors, and management factors—encompassing 32 specific subcategories, as illustrated in Figure 5.

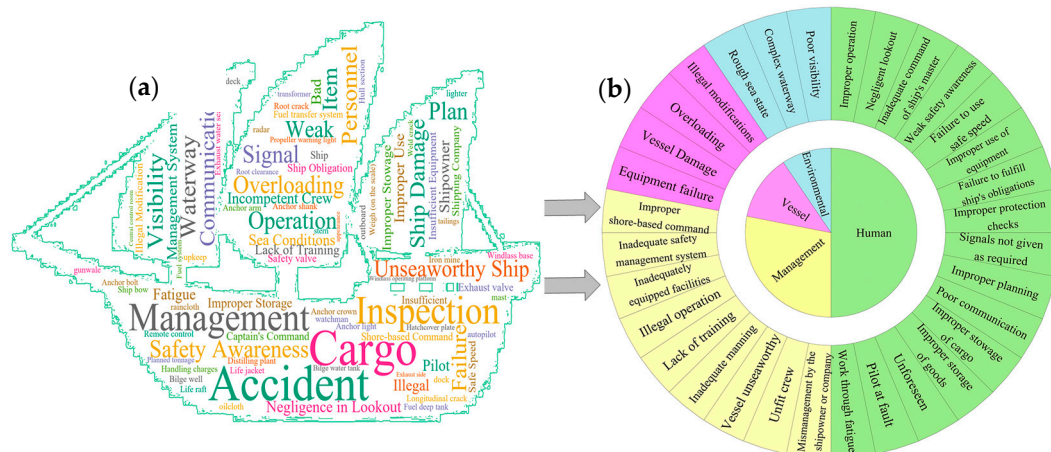


Figure 5. Word cloud diagram (a), Cause classification (b).

Ensuring balanced category representation is crucial for improving model training outcomes [51]. To systematically maintain this balance, this study employs text augmentation techniques, including synonym replacement [52], back translation [53], and even AI generation [54]. These methods expand the dataset and ensure an equitable distribution of data across categories. The model is trained exclusively on international accident reports and the augmented dataset, which together encompass extensive details and enhance both the generalizability and practicality of the model [55]. Finally, the dataset comprises 32 accident-cause classifications, with each category containing approximately 800 base

samples and totaling around 25,930 data points. The data are organized in a structured table, where each record includes a “cause of the accident” field extracted from the accident report and a manually labeled “Cause classification” field. For example, a report stating, “The ship failed to properly display navigation lights while operating at night and did not provide clear operational warnings to nearby vessels,” is categorized under “Signals not given as required”. For model training and evaluation, the dataset is partitioned into training, validation, and test sets in a 70%, 15%, and 15% ratio, respectively.

4.2. Modeling

The experimental environment configuration of this study is shown in Table 2:

Table 2. Environment configuration.

Heading	Configuration Information
Operating system	Windows 11
Programming language	Python 3.9
Experimental platform	Jupyter Notebook
GPU	GeForce RTX4060Ti (8G)
CPU	Inter Core i5-12400F
Deep Learning Framework	PyTorch (CUDA 12.4)

According to the data presented in Table 3, the model’s test accuracy remains consistently high (above 0.88) across different parameter settings—including variations in gradient thresholds, hidden layer sizes, and batch sizes—demonstrating robust insensitivity to parameter fluctuations. Moreover, the introduction of dropout and regularization strategies effectively minimizes the disparity between training and test losses, thereby reducing overfitting risk and further enhancing the model’s robustness and generalization across diverse configurations.

Table 3. Parameter analysis.

Categories	Parameters	Test Loss Rate	Test Accuracy	Difference in Loss
Grads	2.5	0.675	0.885	0.28
	2.75	0.467	0.898	0.26
	3	0.695	0.875	0.43
Hidden layer	64	0.645	0.890	0.49
	128	0.467	0.898	0.26
	256	0.700	0.888	0.68
Batchsize	16	0.708	0.883	0.56
	32	0.467	0.898	0.26
	64	0.648	0.889	0.41
Droupout	0	-	-	0.83
	1	-	-	0.67
	2	-	-	0.26
Regularization	1	-	-	0.54
	2	-	-	0.26

As illustrated in Figure 6, the L2 norm of most gradients is concentrated in the range of 0 to 0.2, exhibiting a pronounced right-skewed long-tail distribution, with only a minor fraction of gradients exceeding 0.3. This distribution indicates that, during training, the network parameters are updated in modest increments, significantly reducing the

likelihood of gradient explosion or related instabilities. Although a few updates exhibit relatively larger gradients, their proportion is negligible based on the histogram.

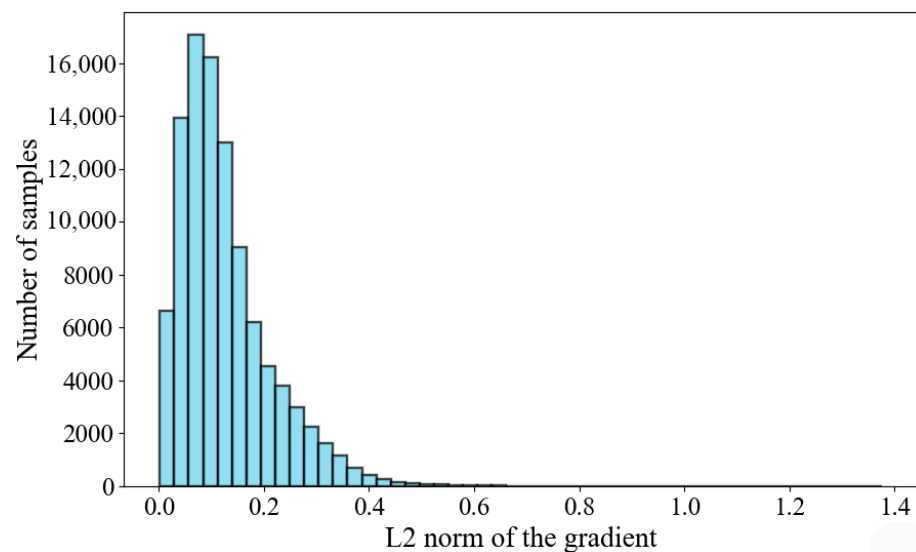


Figure 6. Gradient L2 norm change.

In summary, the convergence behavior observed during training is notably stable, with no significant signs of numerical instability.

4.2.1. Loss Function and Accuracy

In Figure 7a, both training and validation accuracy rise rapidly in the initial phase and plateau at approximately 0.9 after around 20 epochs. Similarly, Figure 7b shows that training and validation loss decrease overall, with higher losses observed at the beginning and a gradual decline leading to stabilization in later stages. This behavior indicates robust classification performance and no apparent overfitting. The model effectively learns and optimizes throughout training, ultimately reaching a stable convergence.

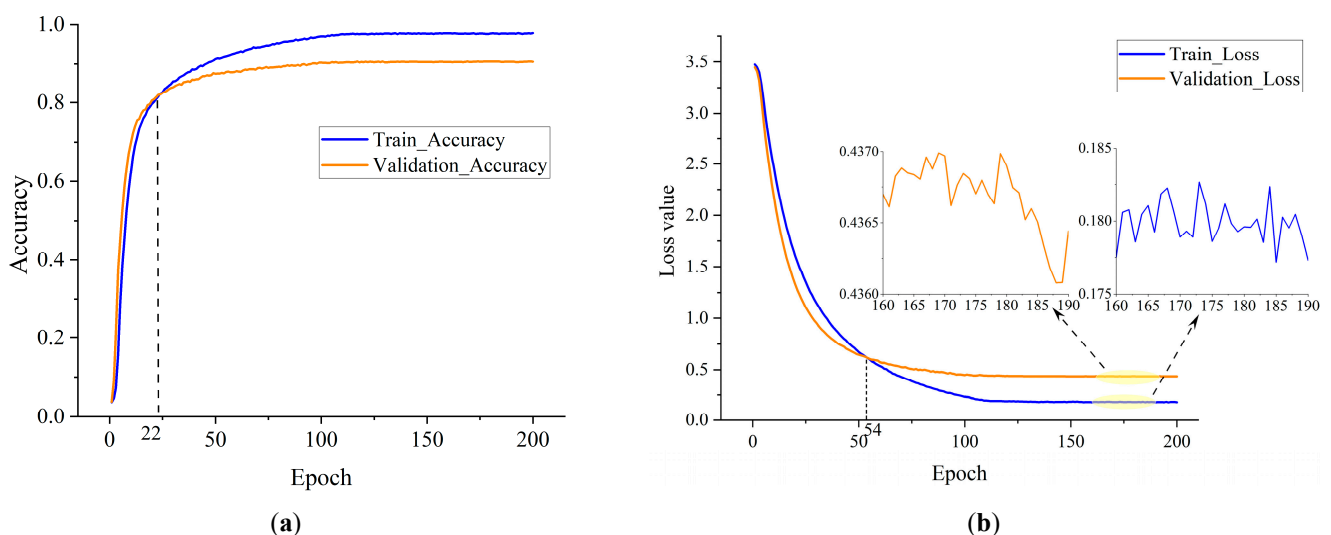


Figure 7. BERT + BiLSTM model accuracy (a) and loss (b).

4.2.2. Confusion Matrix

Figure 8 presents the prediction performance of the BERT + BiLSTM model across various accident cause categories via a confusion matrix. The horizontal axis represents the

predicted category labels, while the vertical axis denotes the true labels. Each cell in the matrix indicates the number of predictions for a specific category. Notably, the diagonal cells are darkened—darker shades correspond to a higher count of correct predictions—whereas off-diagonal cells reveal the frequency and distribution of misclassifications. The pronounced concentration of data along the diagonal implies that the model accurately predicts accident causes across most categories, such as “signals not given as required”. This finding highlights the model’s robustness and reliability in classifying accident causes across a diverse set of categories.

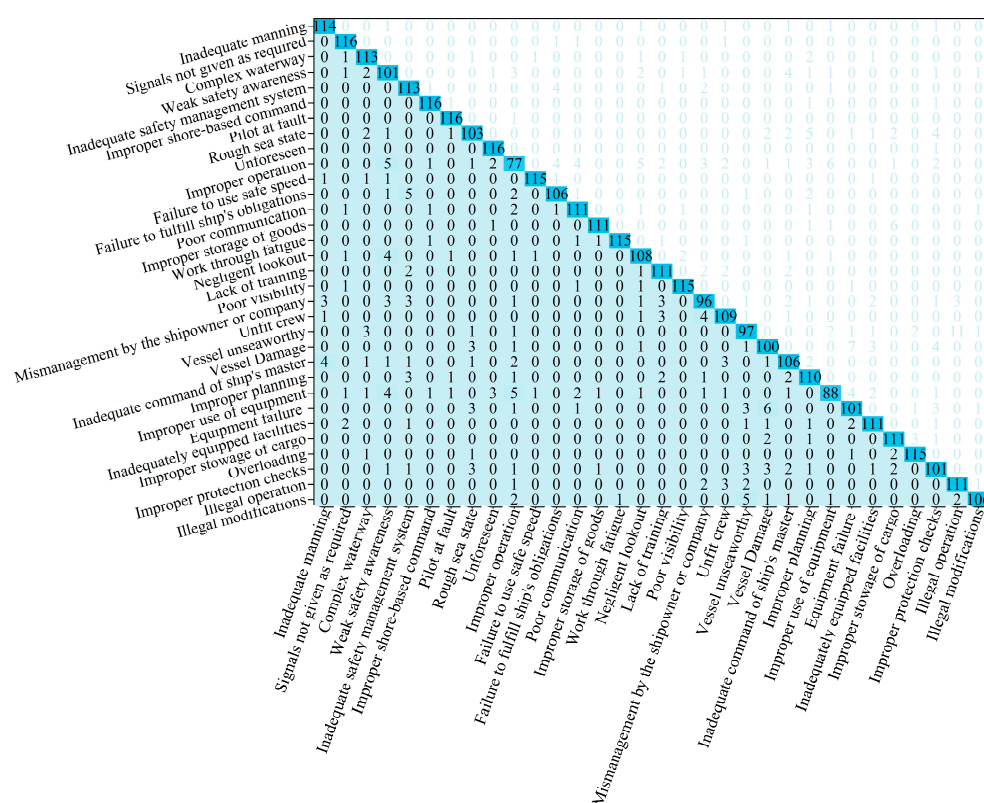


Figure 8. BERT + BiLSTM model confusion matrix.

4.3. Classification Results

The trained BERT + BiLSTM text classification model was used to classify 6268 accident causes in 1337 reports on marine traffic accidents in China. The classification results are shown in Figure 9, and the specific causes are combined with the accident types and levels, as shown in Figure 10.

Figure 9 illustrates the relative importance of various factors in accident causation, with human factors being the most influential. Further analysis reveals that improper operation, inadequate management, and adverse sea conditions are the primary contributors to accidents. Within the ship factors category, elements such as ship damage, overloading, and illegal modifications exert influence under specific circumstances, although their impact is generally less pronounced than that of equipment failure. Among environmental factors, adverse sea conditions dominate, while complex waterways and poor visibility also contribute to accident occurrence. Although issues such as negligence in lookout, fatigue, poor communication, and improper planning are significant within human factors, their influence is somewhat smaller than that of improper operation. At the management level, the findings underscore the need to strengthen the role of shipowners and enhance training initiatives.

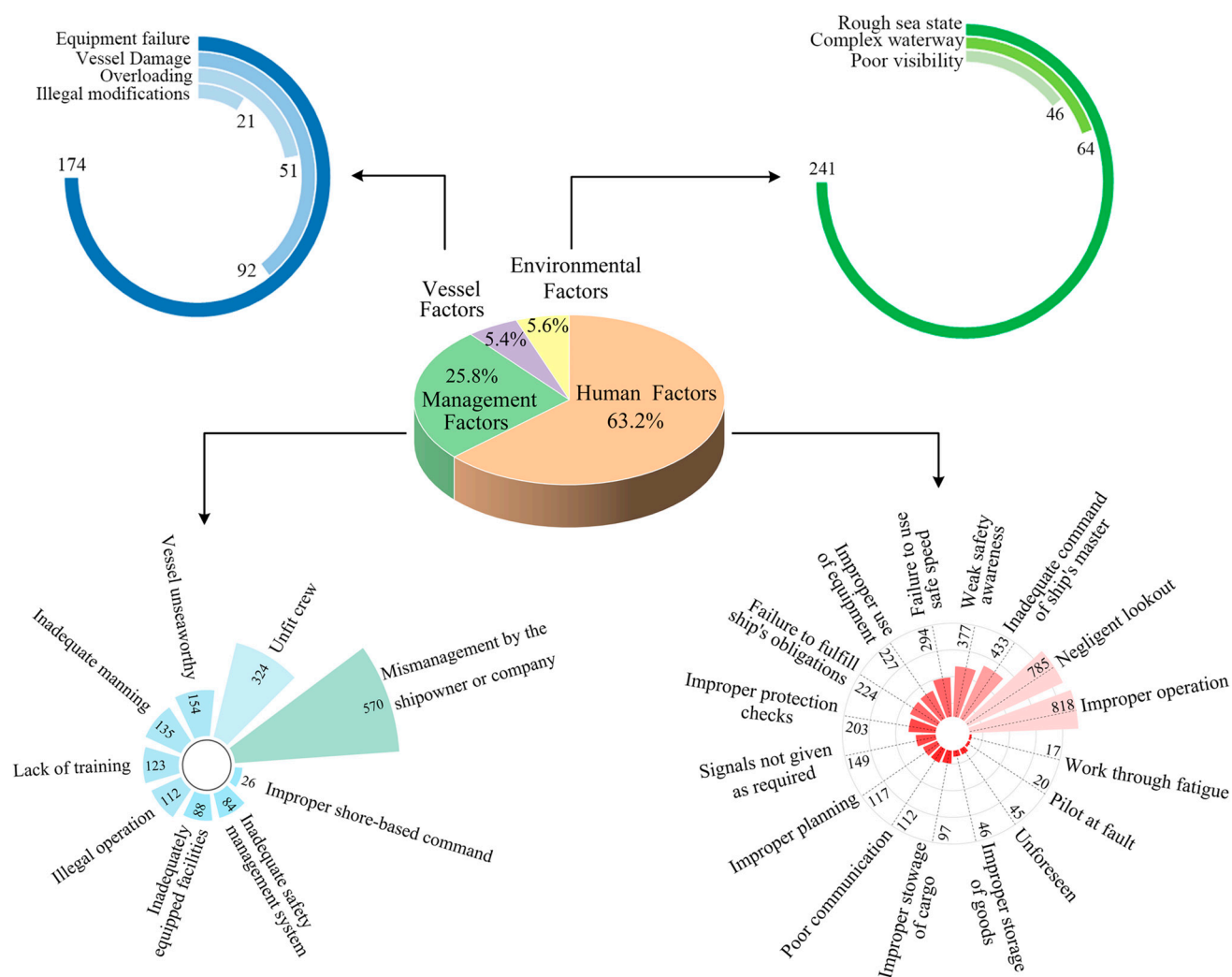


Figure 9. Classification of accident causes.

Figure 10 reveals the inherent logical relationship among accident level, accident type, and specific accident cause. The data show that general-level accidents occur most frequently, whereas major and large-scale accidents are less common. Notably, major and large accidents tend to be closely associated with particular accident types and causes, such as sinking incidents.

In general-level accidents, human factors are predominant. Primary causes include improper operation, negligent lookout, and fatigue, suggesting that most such incidents stem from crew operational errors or negligence. Notably, collision accidents are the most frequent, while contact and grounding accidents, although classified as general-level incidents, are relatively rare. Additionally, factors such as poor communication, failure to signal as required, and inadequate safety management systems also contribute to general accidents, albeit to a lesser extent. For large and major accidents, external environmental factors and critical equipment failures are decisive. Equipment malfunctions, poor management, and deficiencies in safety management systems are frequently linked to severe accident types, such as sinking and fire. In these incidents, equipment failures tend to exacerbate the consequences, while management shortcomings amplify the negative impacts. Furthermore, issues such as overloading and illegal modifications significantly contribute to the severity of these accidents.

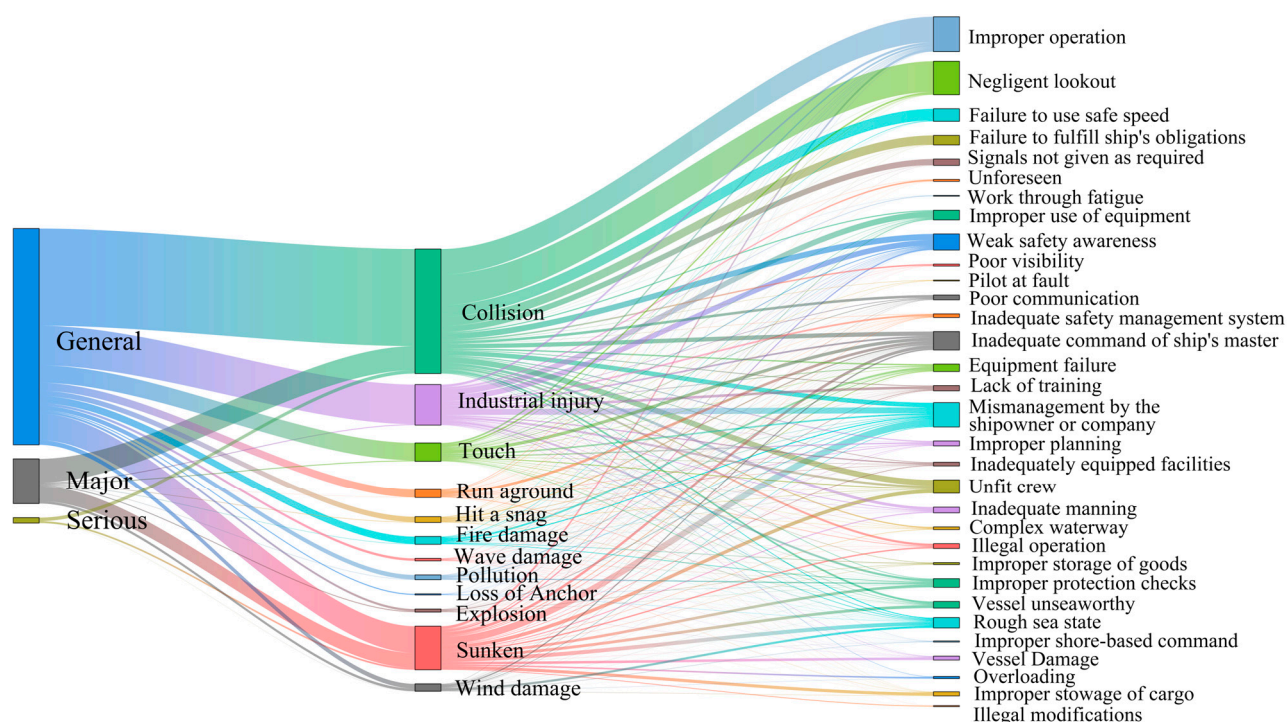


Figure 10. Sankey diagram.

Overall, the occurrence of various accident levels and types is closely linked to specific causal factors. General-level accidents are predominantly influenced by human factors, whereas major and severe accidents are largely driven by equipment failures, management deficiencies, and adverse environmental conditions. Consequently, addressing these incidents requires a comprehensive approach that considers multiple factors rather than focusing on a single aspect. In light of this insight, the Apriori association rule analysis method is employed in this study to further investigate the correlations among accident causes.

4.4. Apriori Association Results

This study employs the output of the BERT + BiLSTM classification model as input for the Apriori algorithm, thereby enhancing the investigation of multi-factor relationships. In the algorithm construction phase (see Figure 11), an increase in the support threshold from 0.005 to 0.02 sharply reduces the number of generated rules—from approximately 450 to about 50—with the rate of reduction beginning to level off at a support value around 0.008. Similarly, as the confidence threshold increases from 0.05 to 0.25, the number of rules decreases from roughly 700 to 180, with a notable inflection occurring at approximately 0.15. Based on these observations and iterative testing, the threshold for accident association rules is set to $Min_support \geq 0.008$ and $Min_confidence \geq 0.15$ to balance rule quality and coverage. Furthermore, to construct a comprehensive causal chain and capture potential associations, the thresholds for the accident causal chain are established at $Min_support \geq 0.01$ and $Min_confidence \geq 0.1$.

As shown in Figure 12, the Apriori algorithm processes accident causes derived from data handled by the BERT + BiLSTM model. Initially, raw data—including serial numbers and classification fields—are input into the system. During preprocessing, classification labels are aggregated by serial number, duplicates are removed, and the labels are transformed into a Boolean matrix. Subsequently, the Apriori algorithm identifies high-frequency item sets and generates association rules, eliminating redundant subsets before outputting the final rules.

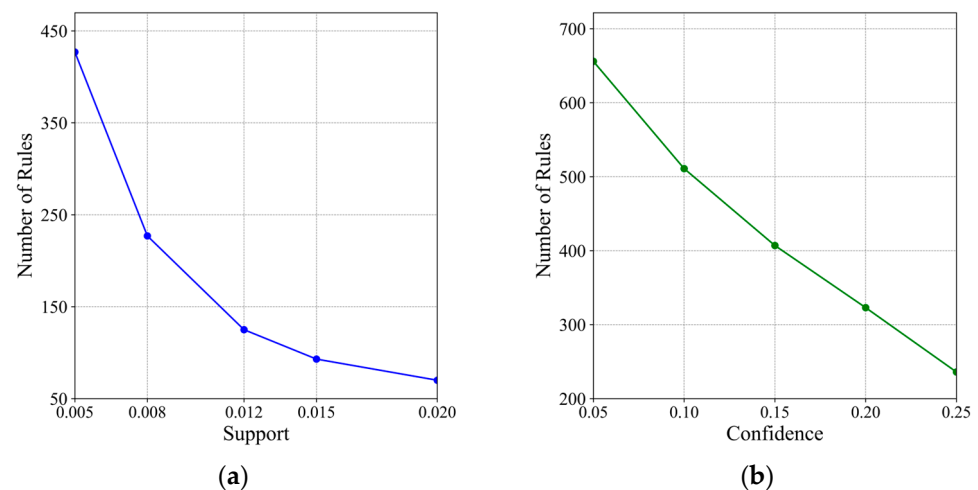


Figure 11. Association rule threshold selection: changes in support (a) and changes in confidence (b).

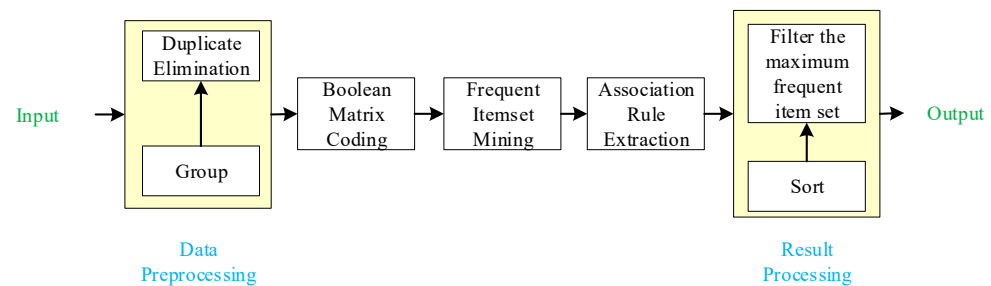


Figure 12. Function of association rule module.

By analyzing ship accident data, critical factors and their relationships were identified and then compared, analyzed, and summarized based on single-factor and multi-factor criteria:

- (1) The most widespread causes

The high support in Table 4 shows that these factors are pivotal in accident occurrence and represent key areas for improvement. Moreover, a higher node centrality implies more associations with other accident causes, suggesting that such factors may significantly influence multiple aspects of accident causation.

Table 4. Degree centrality ranking.

Sort Number	The Specific Cause of the Accident	Degree Centrality
1	Mismanagement by the shipowner or company	0.9677
2	Improper operation	0.8710
3	Weak safety awareness	0.8710
4	Unfit crew	0.8387
5	Inadequate command of the ship's master	0.8387
6	Rough sea state	0.6774
7	Inadequate manning	0.6452
8	Vessel unseaworthy	0.6129
9	Poor communication	0.6129
10	Negligent lookout	0.6129

- (2) Sorting of strong association rules

The high-confidence rules indicate that under specific preconditions, the likelihood of a particular outcome is significantly elevated. As demonstrated by the rules in Tables 5 and 6—sorted by confidence—it becomes imperative to enhance training programs and upgrade equipment to improve operational standardization and personnel safety awareness. Simultaneously, the development of a multi-factor risk assessment system is crucial. Regular safety assessments should be implemented, particularly focusing on critical combinations, such as “Failure to use a safe speed”, “Inadequate manning” and “Negligent lookout”. For risks arising from the interplay of multiple factors, a systematic prevention strategy must be formulated, and potential hazards should be continuously monitored and addressed in real time through a dedicated early warning system.

Table 5. Confidence of two-factor correlation.

Sort Number	Cause Combination	Lift	Confidence	Support
1	Failure to use safe speed -> Negligent lookout	2.23	0.81	0.13
2	Signals not given as required -> Negligent lookout	2.22	0.80	0.07
3	Failure to fulfill ship’s obligations -> Negligent lookout	2.13	0.77	0.09
4	Pilot at fault -> Inadequate command of the ship’s master	3.49	0.76	0.01
5	Illegal modifications -> Mismanagement by the shipowner or company	2.45	0.76	0.01

Table 6. Confidence of multi-factor correlation.

Sort Number	Cause Combination	Lift	Confidence	Support
1	Inadequate manning, Improper operation, Failure to use safe speed -> Negligent lookout	2.77	1	0.01
2	Vessel unseaworthy, Signals not given as required -> Improper operation	2.33	0.92	0.01
3	Poor communication, Failure to fulfill ship’s obligations -> Improper operation	2.33	0.92	0.01
4	Pilot at fault, Improper operation -> Inadequate command of ship’s master	4.1	0.90	0.01
5	Signals not given as required, Failure to use safe speed -> Negligent lookout	2.47	0.89	0.03

The high-impact improvement rule demonstrates that both individual factors and their combinations substantially elevate risk, warranting prioritized management interventions. This is reflected in Tables 7 and 8. For example, “Work through fatigue” increases the risk of “Improper use of equipment”, suggesting that crew work schedules should be planned to mitigate fatigue. Similarly, “Illegal modifications” lead to “Vessel unseaworthiness”, indicating the need for enhanced supervision of modification practices and strict seaworthiness inspections. Furthermore, the combination of “Inadequate command of the ship’s master” and “Improper operation” elevates the risk of “Pilot at fault”, underscoring the importance of comprehensive management, improved training, and rigorous supervision. Additionally, the joint occurrence of “Vessel unseaworthy” and “Improper operation” heightens the risks of “Signals not given as required” and “Unfit crew”, necessitating systematic preventive measures to ensure both vessel integrity and crew qualifications.

Based on confidence levels and degrees of enhancement, both single-factor and multi-factor scenarios are ranked, offering a comprehensive illustration of the diverse factors influencing accidents.

Table 7. Lift of two-factor correlation.

Sort Number	Cause Combination	Lift	Confidence	Support
1	Work through fatigue -> Improper use of equipment	4.29	0.53	0.01
2	Illegal modifications -> Vessel unseaworthy	3.73	0.37	0.01
3	Inadequate safety management system -> Lack of training	3.50	0.27	0.02
4	Lack of training -> Inadequate safety management system	3.50	0.20	0.02
5	Pilot at fault -> Inadequate command of the ship's master	3.49	0.76	0.01

Table 8. Lift of multi-factor correlation.

Sort Number	Cause Combination	Lift	Confidence	Support
1	Inadequate command of ship's master, Improper operation -> Pilot at fault	9.07	0.15	0.01
2	Pilot at fault -> Inadequate command of ship's master, Improper operation	9.07	0.53	0.01
3	Vessel unseaworthy, Improper operation -> Signals not given as required, Unfit crew	8.56	0.17	0.01
4	Signals not given as required, Unfit crew -> Vessel unseaworthy, Improper operation	8.56	0.27	0.01
5	Signals not given as required, Vessel unseaworthy -> Unfit crew, Improper operation	7.95	0.58	0.01

5. Discussion

5.1. Model Comparison

This paper compares the loss function, accuracy, and confusion matrix among three models: BERT alone, BERT with a pooling mechanism, and BERT + BiLSTM.

In this study (Table 9), all three models are configured with the same learning rate (1×10^{-6}), batch size (32), and AdamW optimizer; however, there are targeted differences in their network architectures and regularization strategies. Specifically, both BERT and BERT + Pooling Mechanism directly utilize BERT's 512-dimensional hidden layer, whereas BERT + BiLSTM incorporates an additional 128-unit BiLSTM layer to capture sequential information more effectively. Furthermore, the models differ in their dropout settings, L1 and L2 regularization coefficients, and gradient constraints. Regarding activation functions, BERT employs GELU, while the other models use Mish to enhance their nonlinear expression capabilities.

Table 9. Model parameter settings.

Parameter	BERT	Pooling Mechanism + BERT	BERT + BiLSTM
Hidden Layer	BERT layer 512	BERT layer 512	LSTM layer 128
Learning Rate	1×10^{-6}	1×10^{-6}	1×10^{-6}
Dropout Layer	3	3	2
L1 Regularization	1×10^{-8}	1×10^{-8}	5×10^{-10}
L2 Regularization	0.05	0.01	0.05
Gradient	2.35	3.5	2.75
Optimizer	AdamW	AdamW	AdamW
Activation Function	GELU	Mish	Mish
Batchsize	32	32	32

Overall, the fine-tuning of these hyperparameters is designed to balance model stability with the capacity to capture complex textual sequence information, thereby optimizing classification performance.

In the three sets of graphs (Figures 7, 13 and 14), the BERT + BiLSTM model demonstrates rapid convergence, achieving high training and validation accuracy within approximately 20 epochs and maintaining stable performance thereafter. Its loss curve remains consistent, indicating robust generalization. By contrast, while the BERT model shows early improvements, its validation loss begins to fluctuate markedly after around 50 epochs, suggesting reduced stability. The BERT + Pooling Mechanism model exhibits similar performance in terms of accuracy compared to the other two models; however, its loss function converges at a slower rate and demonstrates greater fluctuation in the later stages of training. Overall, the BERT + BiLSTM model exhibits clear advantages in both convergence speed and final accuracy.

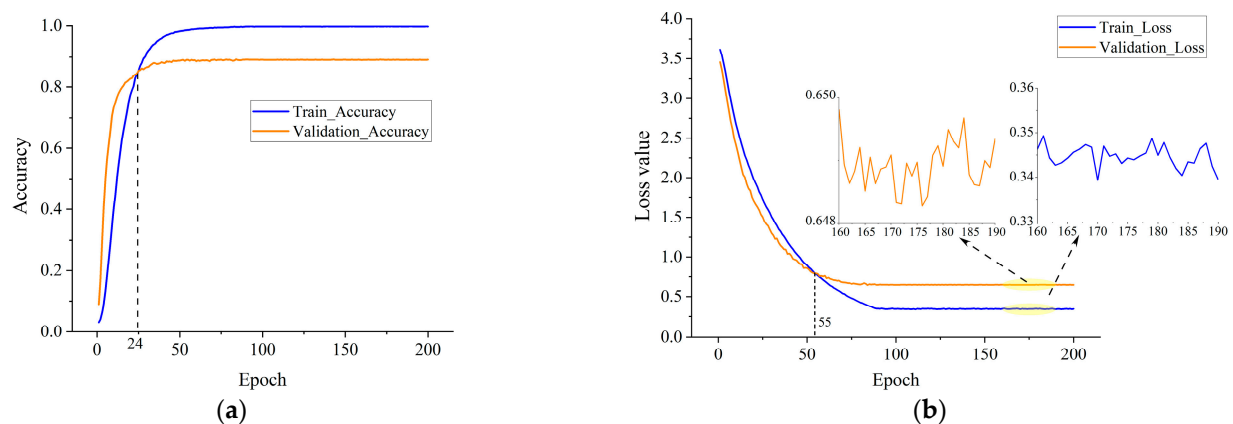


Figure 13. BERT model accuracy (a) and loss (b).

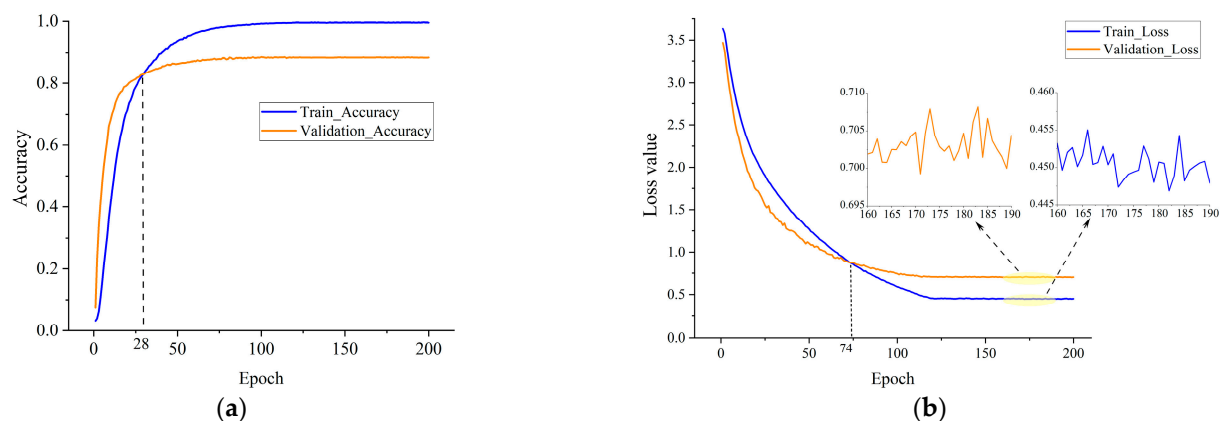


Figure 14. BERT+ Pooling Mechanism model accuracy (a) and loss (b).

An examination of the three confusion matrices reveals that the BERT + BiLSTM model (Figure 8) consistently achieves higher diagonal values across most categories, reflecting superior classification accuracy. In contrast, the confusion matrix for the BERT model (Figure 15a) displays a more dispersed diagonal, with a higher rate of misclassifications in certain categories, which suggests limited discriminative power among similar classes. The BERT + Pooling Mechanism model (Figure 15b) demonstrates relatively high accuracy along the diagonal; however, it still exhibits some confusion in specific categories compared to the BERT + BiLSTM model. Overall, the BERT + BiLSTM model outperforms the other models in both classification accuracy and stability. While the BERT + Pooling Mechanism

model does enhance the performance of the original BERT model to some extent, it remains slightly inferior to the approach incorporating bidirectional LSTM.

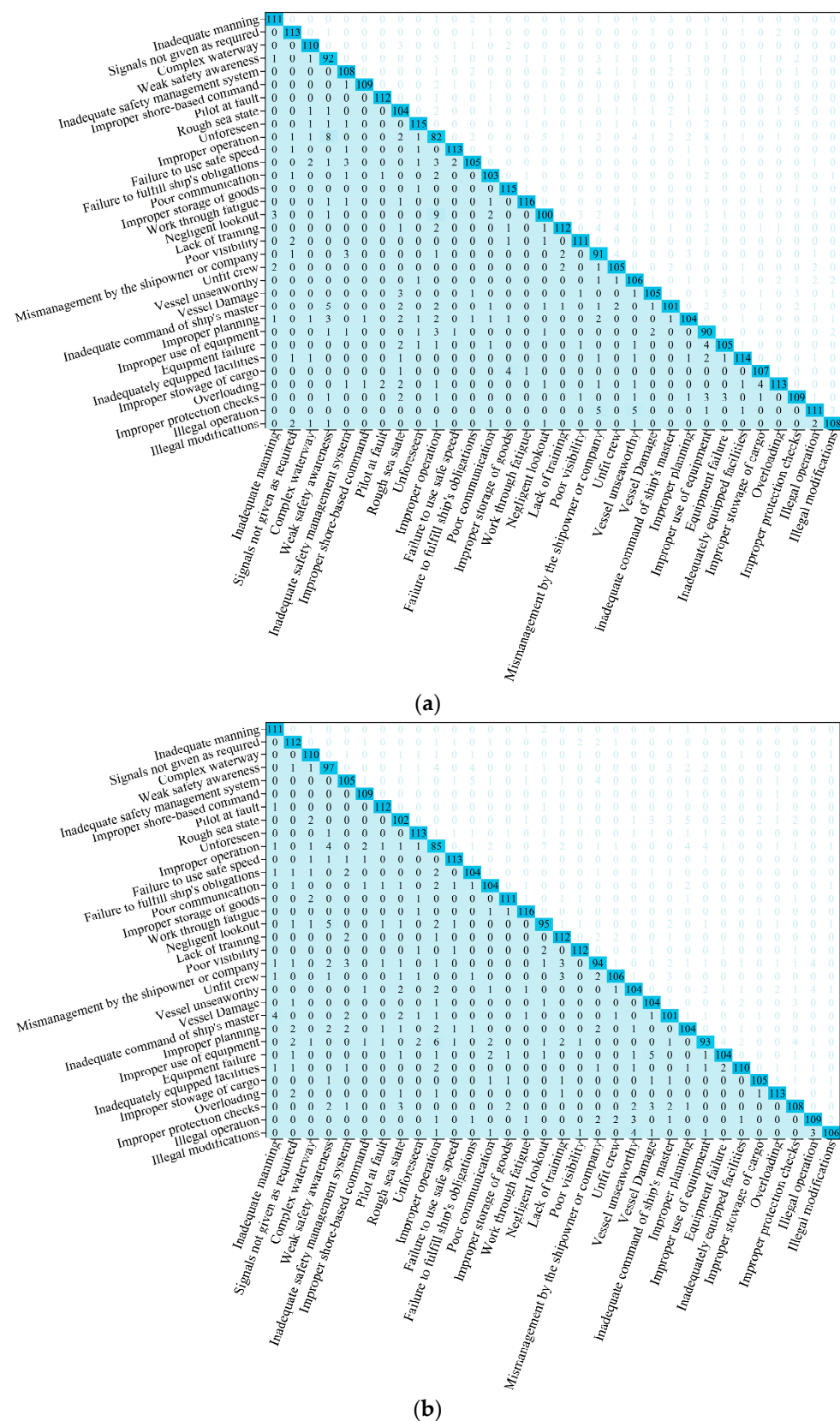


Figure 15. Comparison of obfuscation matrices for BERT (a) and pooling mechanisms (b).

Table 10 presents the evaluation of the model’s accident cause classification performance using precision, recall, and F1-score. Here, precision reflects the frequency with which the model correctly predicts a specific category, while recall indicates the proportion of correctly identified samples within that category relative to all actual samples. The F1-score, serving as the harmonic mean of precision and recall, offers a balanced measure of accuracy and coverage. Statistically, the BERT + BiLSTM model achieves high F1 scores across most accident cause categories. It performs particularly well in critical categories, such as “Signals not given as required”, “Pilot at fault”, and “Work through fatigue”, demonstrating its robustness in handling complex text sequences and multi-class classification tasks. In contrast, the standalone BERT model shows significant fluctuations in certain categories (e.g., “Improper operation”), and while the BERT + Pooling Mechanism model improves classification performance relative to BERT, it remains slightly inferior to the BERT + BiLSTM approach.

Table 10. Model indicator comparison.

Cause	Model Evaluation Indicators	BERT			BERT + BiLSTM			BERT + Pooling Mechanism		
		Precision	Recall	F1-Score	Precision	Recall	F1-Score	Precision	Recall	F1-Score
Inadequate manning		0.925	0.941	0.933	0.927	0.950	0.938	0.917	0.941	0.929
Signals not given as required		0.934	0.926	0.930	0.944	0.975	0.959	0.889	0.918	0.903
Complex waterway		0.932	0.924	0.928	0.911	0.942	0.926	0.909	0.924	0.917
Weak safety awareness		0.821	0.786	0.803	0.821	0.842	0.831	0.836	0.829	0.833
Inadequate safety management system		0.857	0.893	0.874	0.876	0.934	0.904	0.875	0.868	0.871
Improper shore-based command		0.940	0.982	0.960	0.967	0.983	0.975	0.932	0.982	0.956
Pilot at fault		0.949	0.974	0.961	0.975	0.992	0.983	0.949	0.974	0.961
Rough sea state		0.846	0.825	0.835	0.880	0.837	0.858	0.857	0.810	0.833
Unforeseen		0.927	0.935	0.931	0.943	0.991	0.967	0.911	0.919	0.915
Improper operation		0.656	0.672	0.664	0.755	0.642	0.694	0.720	0.697	0.708
Failure to use safe speed		0.942	0.950	0.946	0.975	0.958	0.966	0.934	0.950	0.942
Failure to fulfill ship’s obligations		0.882	0.913	0.897	0.906	0.898	0.902	0.881	0.904	0.893
Poor communication		0.920	0.888	0.904	0.910	0.925	0.917	0.889	0.897	0.893
Improper storage of goods		0.958	0.927	0.943	0.974	0.925	0.949	0.949	0.895	0.921
Work through fatigue		0.951	0.959	0.955	0.991	0.958	0.975	0.936	0.967	0.951
Negligent lookout		0.813	0.877	0.844	0.871	0.900	0.885	0.841	0.833	0.837
Lack of training		0.862	0.933	0.896	0.902	0.925	0.914	0.875	0.933	0.903
Poor visibility		0.957	0.933	0.945	0.958	0.966	0.962	0.933	0.941	0.937
Mismanagement by the shipowner or company		0.858	0.758	0.805	0.857	0.800	0.828	0.847	0.783	0.814
Unfit crew		0.913	0.875	0.894	0.893	0.908	0.901	0.922	0.883	0.902
Vessel unseaworthy		0.891	0.855	0.872	0.836	0.808	0.822	0.874	0.839	0.856
Vessel Damage		0.854	0.882	0.868	0.855	0.833	0.844	0.860	0.874	0.867
Inadequate command of the ship’s master		0.821	0.863	0.842	0.841	0.869	0.855	0.821	0.863	0.842
Improper planning		0.846	0.867	0.856	0.833	0.917	0.873	0.825	0.867	0.846
Improper use of equipment		0.891	0.720	0.796	0.863	0.739	0.796	0.845	0.744	0.791
Equipment failure		0.861	0.882	0.871	0.856	0.842	0.849	0.860	0.874	0.867
Inadequately equipped facilities		0.919	0.919	0.919	0.902	0.925	0.914	0.924	0.887	0.905
Improper stowage of cargo		0.939	0.907	0.922	0.895	0.941	0.917	0.905	0.890	0.897
Overloading		0.883	0.934	0.908	0.950	0.958	0.954	0.883	0.934	0.908
Improper protection checks		0.858	0.879	0.869	0.828	0.842	0.835	0.850	0.871	0.861
Illegal operation		0.881	0.902	0.892	0.874	0.925	0.899	0.893	0.886	0.890
Illegal modifications		0.915	0.915	0.915	0.955	0.891	0.922	0.906	0.898	0.902
Accuracy		0.887	0.887	0.887	0.898	0.898	0.898	0.883	0.883	0.883
Macro avg		0.888	0.887	0.887	0.898	0.898	0.897	0.883	0.884	0.883
Weighted avg		0.887	0.887	0.886	0.898	0.898	0.897	0.883	0.883	0.882

Overall, in terms of overall accuracy, macro-average, and weighted-average metrics, the BERT + BiLSTM model leads with an overall performance of approximately 0.898, underscoring its clear advantages in capturing textual sequence information and managing multi-class classification challenges.

5.2. Apriori Algorithm Analysis

5.2.1. Complex Network Analysis

Figure 16, based on edge importance from the Apriori association rule analysis, illustrates a complex network of maritime accident causes. As depicted in the figure, the factors are not independent but are intricately interconnected, forming a highly integrated and

5.2.2. Accident Causation Chain

Guided by the accident type, the Apriori association rule analysis of the accident causes was performed. After multiple iterations, the threshold selection range was $Min_support \geq 0.01$ and $Min_confidence \geq 0.1$. During the construction of the accident correlation network, unreasonable correlation paths were systematically excluded to ensure the reliability of the analysis. Based on the algorithmic results, we identified and summarized the causal chains for different types of accidents, which are illustrated by Figure 17 (collision accidents), Figure 18 (sinking accidents), and Figure 19 (pollution accidents). Figure 17 demonstrates that in collision accidents, complex interactions among human errors, equipment malfunctions, and management oversights lead to the occurrence of accidents. Figure 18 elucidates the causal chain for sinking accidents, revealing the influence of adverse sea conditions, equipment failures, and other contributing factors. Meanwhile, Figure 19 highlights the synergistic effects of multiple factors—including environmental conditions, communication breakdowns, and management deficiencies—in the occurrence of pollution accidents. In general, this causal chain analysis clarifies that the primary vulnerabilities in ship safety management stem from the intricate interplay among various factors, such as personnel, equipment, management, and communication. While a single factor is generally insufficient to independently cause an accident, the convergence of these factors significantly elevates the risk of accidents.

(1) Collision

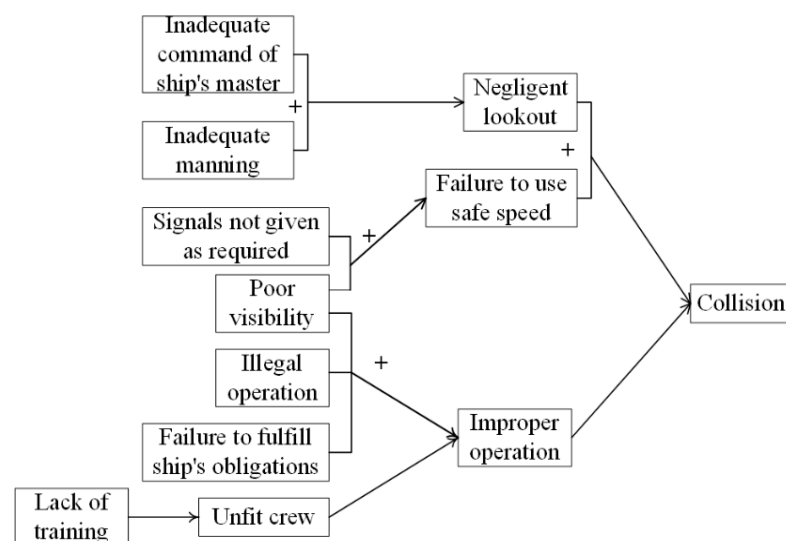


Figure 17. Collision accident chain.

(2) Sunken

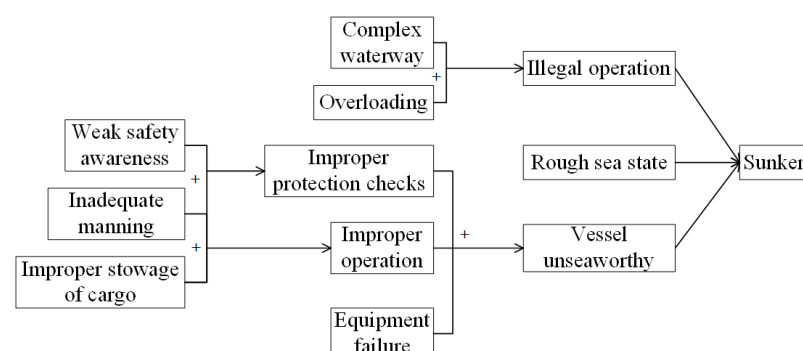


Figure 18. Sunken accident chain.

(3) Pollution

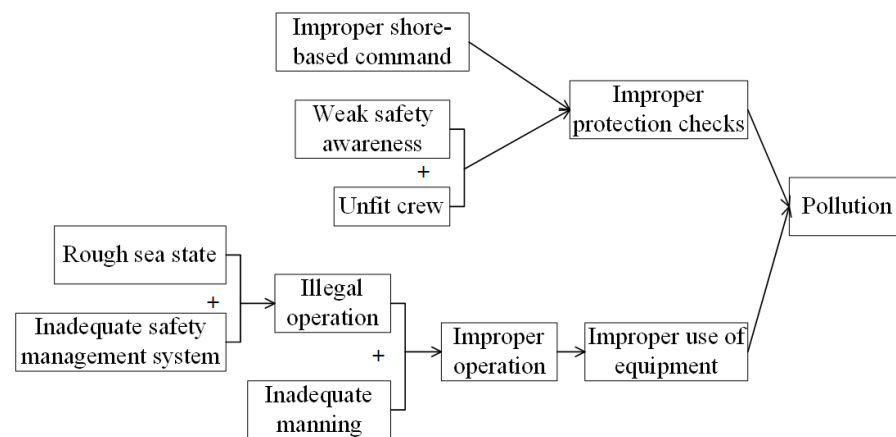


Figure 19. Pollution accident chain.

5.3. Analysis of Accident Information

This section is based on the statistical analysis of accident reports, with a focus on differences in accident levels, spatial distribution, accident types, and seasonal characteristics. The goal is to uncover the underlying patterns in accident occurrence.

5.3.1. Accident Level and Areas

According to the accident classification standards issued by the China Maritime Safety Administration, there were 1132 general accidents, 187 major accidents, and 18 serious accidents, as detailed in Table 11. Analysis of published accident reports indicates that water transport accidents in China predominantly fall under the category of general accidents, which tend to result in relatively few casualties (missing persons).

Table 11. Accident classification criteria.

Accident Level	Number of Deaths (Missing)	Number of Injured	Direct Economic Losses
Extraordinary major accidents	More than 30 people	More than 100 people	More than 100 million RMB
Serious accidents	More than 10 people but less than 30 people	More than 50 people but less than 100 people	More than 50 million RMB and less than 100 million RMB
Major accident	More than 3 people but less than 10 people	More than 10 people but less than 50 people	More than 10 million RMB and less than 50 million RMB
General accidents	More than 1 person but less than 3 people	More than 1 person but less than 10 people	Less than 10 million RMB

Figure 20 shows the distribution of accident levels in mainland China during daytime (06:00–18:00) and nighttime (18:01–06:00). A comparison of the density maps reveals that both the frequency and severity of accidents increase at night. In particular, the locations of large and major accidents are notably denser on the nighttime map. Additionally, accidents at night tend to cluster more in specific areas, especially along coastal and urban regions, which may be attributed to factors such as altered driving conditions and reduced visibility in low-light environments.

In the meantime, a heat map of coastal marine traffic accidents in China was drawn based on the accident locations, as shown in Figure 21 (The deeper the red hue, the greater the number of accidents reported). The main locations are the Bohai Sea Rim, the Yangtze River Estuary, and the Pearl River Delta. This hot spot distribution is also in line with the rule of ship traffic.

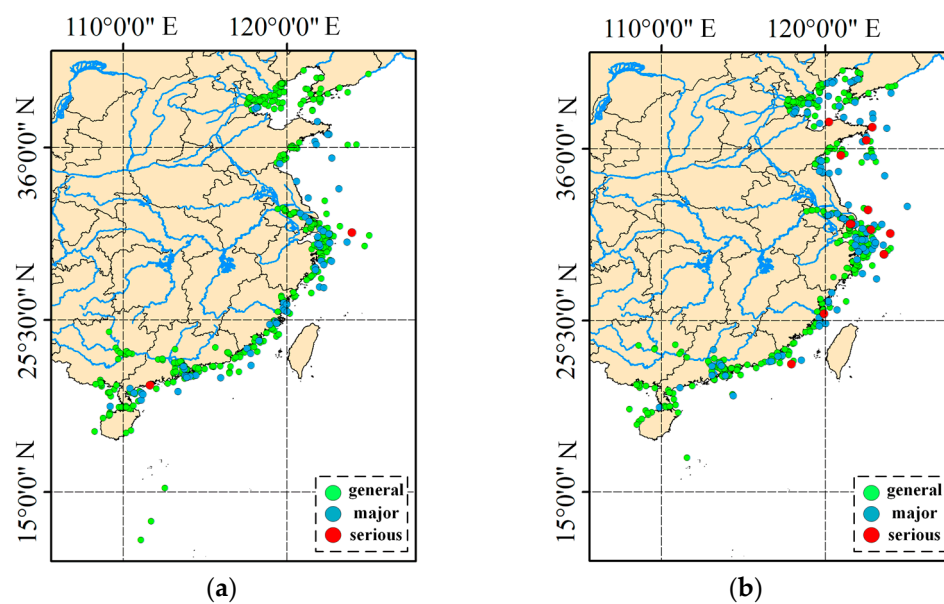


Figure 20. Accident level distribution: day (a) and night (b).

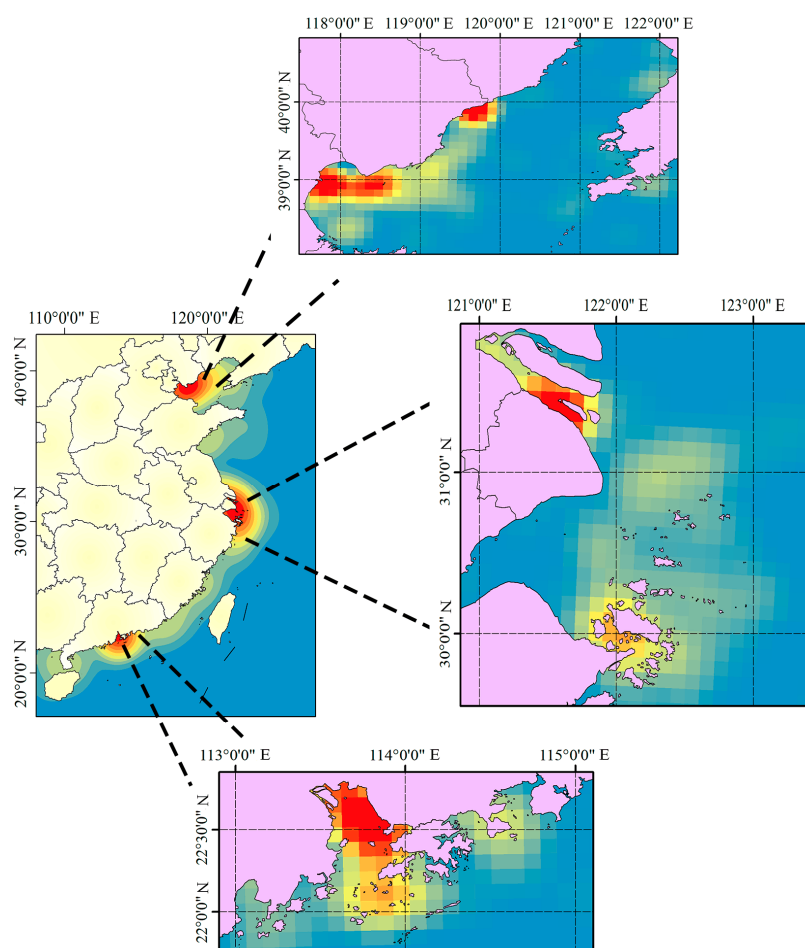


Figure 21. Heat map of accident.

5.3.2. Type of Accident

Figure 22 presents the distribution of over ten accident types, with collision accidents being the most prevalent at 544 cases, followed by industrial injuries (219 cases) and sinking accidents (217 cases). Other accident types, such as fire or explosion and grounding, occur relatively infrequently. Notably, collision and sinking accidents not only dominate in

frequency but also have a disproportionate impact on casualty numbers. Collision accidents have resulted in over 700 casualties, far exceeding the nearly 500 casualties from sinking accidents. These findings highlight the critical need to prioritize collision and sinking accident prevention in maritime safety management.

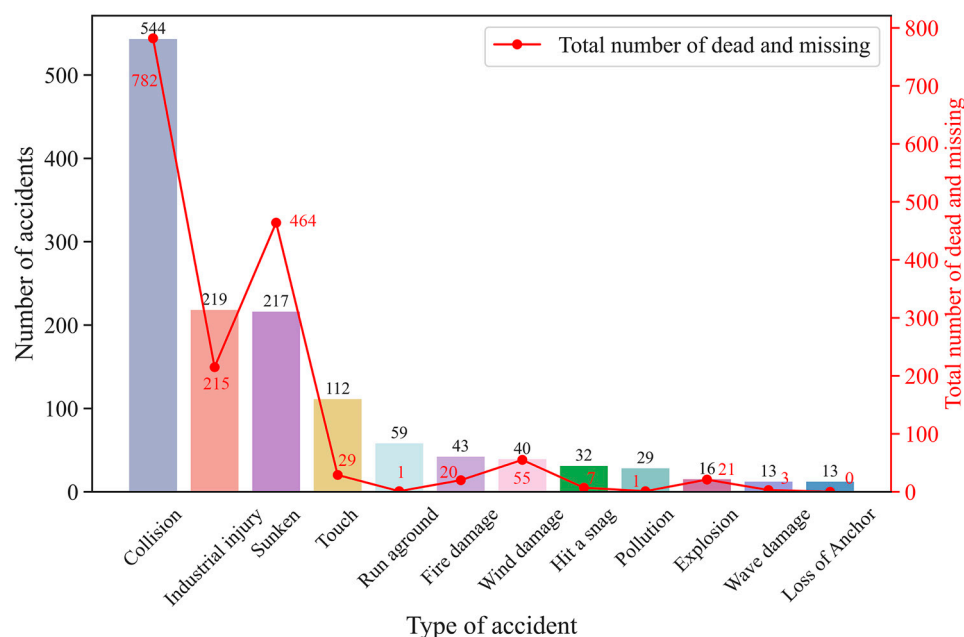


Figure 22. Types of accidents and statistics on the number of dead and missing.

Furthermore, the figure highlights an important statistical trend: Despite the relatively low frequency of fire and explosion accidents, the fatalities and disappearances per incident are notably high. This suggests that such accidents, although less common, pose significant inherent risks and require targeted preventive measures. Similarly, while grounding accidents occur more frequently, the casualty rates associated with them are comparatively low, potentially reflecting the effectiveness of current response strategies in mitigating their impact.

5.3.3. Seasonal Influences

Previous studies have largely overlooked the influence of seasonal variations on maritime accidents. In this paper, as shown in Figure 23, seasonal factors significantly impact both the frequency and type of accidents. In particular, severe winter weather conditions markedly increase the likelihood of explosion and sinking accidents. This finding underscores the need to intensify maintenance and inspections of ship equipment—especially fuel and power systems—during the winter months. Moreover, issuing timely warnings and implementing protective measures in response to severe weather conditions are essential strategies to mitigate risks.

Spring and summer constitute peak shipping seasons, during which the frequency of work-related injuries, groundings, and pollution accidents increases significantly. It is advisable to intensify crew safety training during these periods to improve safety awareness and operational standards under high-intensity working conditions. Concurrently, enhanced monitoring and management of waterways and ports are essential to mitigate the risks of groundings and pollution incidents.

In contrast, collision accidents peak during summer and autumn due to heightened ship activities and increased shipping density. Consequently, it is imperative to reinforce traffic management during these seasons, optimize route design, and employ radar and Automatic Identification Systems (AIS) to bolster collision avoidance capabilities.

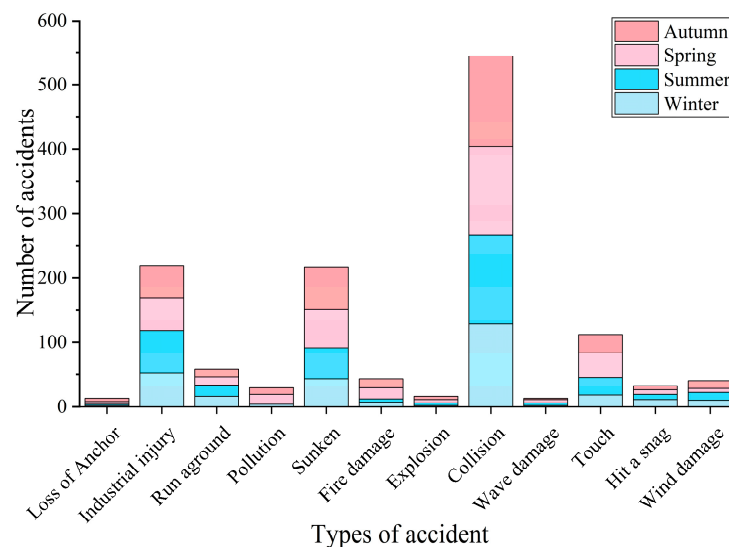


Figure 23. Seasonal chart.

Moreover, fire and wave damage accidents occur year-round, underscoring their strong association with the condition of ship equipment and adherence to operational procedures. Regular inspections and maintenance of ship equipment, coupled with strict enforcement of operating protocols and regulatory compliance among crew members, are therefore recommended throughout the year.

5.4. Improvements

Maritime accident reports serve as a valuable data source in this study due to their status as official documents, enabling a systematic analysis that can effectively mitigate maritime accident risks [56,57]. Table 12 clearly outlines the enhancements introduced in this research.

This paper employs an enhanced BERT model to analyze marine traffic accidents, demonstrating the application of NLP in maritime safety. The multifactor analysis confirms the predominant role of human factors and emphasizes the significance of their interactions. High-confidence association rules—such as the link between fatigue and equipment misuse—are identified, forming the basis for targeted prevention strategies. Moreover, by integrating accident trend analyses with heat maps, this study proposes seasonal management strategies, including upgrading equipment in winter and optimizing routes during the summer/autumn periods.

In comparison to traditional approaches, this research offers substantial improvements in data mining, semantic analysis, and multifactorial investigation, providing novel insights into the processing of unstructured text in maritime safety contexts.

In terms of prevention policy, this paper advocates a multi-level approach. First, based on human factor standards, it recommends enhancing crew training through a classification-based training and certification system. Additionally, improvements should be made to the working environment by enforcing an anti-fatigue system and establishing dual-supervision mechanisms for mental health screening and behavior recording. The development of stringent operating procedures and comprehensive emergency response plans is expected to significantly reduce accidents caused by operational errors or improper judgment.

Table 12. Comparison with existing studies.

Research Topics	Traditional Methods/Current Situation	Improvement Method (This Article)	Comparison and Improvement
Data processing	Quantitative risk analysis [58], relies on structured data and preset parameters [59].	Enhancing the BERT model to directly extract deep semantic information from unstructured accident reports.	BERT offers a profound understanding of textual data, particularly in capturing complex causal relationships, thereby significantly enhancing the analysis of unstructured text [43,55].
Classification of accident causes	Most studies employ structural rule analysis techniques, resulting in findings that tend to be relatively generalized.	Automatic classification is achieved using deep learning models that address four primary factors—human, management, ship, and environment—encompassing a total of 32 subcategories.	The classification granularity is refined, and when combined with data enhancement techniques, the model's coverage and accuracy are improved, thereby providing more targeted data support for accident prevention measures [6].
Accident cause analysis	Previous studies have predominantly emphasized human error as the primary cause of maritime accidents [7,60]. Most research relies on single-factor analysis and employs Bayesian network analysis for risk assessment; however, these approaches lack a comprehensive multi-factor perspective [12,23,61].	Building on the BERT model, accident cause classification is refined and integrated with the Apriori association rule algorithm, thereby revealing the complex interactions among multiple factors and generating high-confidence association rules.	The in-depth analysis of human factors was further refined, emphasizing the critical role of safety awareness [62]. Additionally, a range of integrated solutions were proposed to address combined issues, such as the interplay between fatigue and equipment failure, as well as the link between illegal modifications and ship unseaworthiness [63,64].
Accident trend research	Based on simple distribution analyses, research on the differences between accident types and regions remains limited [6].	Visualization analysis—including heat maps and seasonal trend maps—reveals the distribution patterns of accident types, temporal trends, and regional characteristics.	Nighttime accidents are predominantly concentrated in coastal areas. Shipwrecks occur more frequently during winter, whereas collisions are more common in summer and autumn. These observations have led to the proposal of management recommendations that are both time-based and area-specific [16,65].
Innovation and advantages	Traditional risk analysis methods, such as Bayesian networks and FSA, provide limited support for association rule generation, which hampers the capacity for dynamic analysis [21,58].	The improved BERT model delivers deep semantic understanding and enables the automatic classification of diverse accident causes.	This paper excels in dynamic data mining, effectively integrating it with semantic analysis to substantially enhance the comprehensive management of multiple factors.

Second, at the technical prevention and control level, the establishment of regular maintenance and preventive maintenance systems is essential for addressing non-human-related causes. Real-time monitoring and predictive maintenance methodologies should be employed to promptly identify and eliminate potential equipment hazards. Legislative measures could mandate the installation of intelligent sensing systems on ships, and a national ship retrofitting record along with a dynamic sampling inspection mechanism could be implemented to further safeguard maritime operations.

Finally, at the system collaboration level, a dynamic risk assessment platform involving maritime departments, shipping enterprises, and insurance institutions is recommended. By promoting data sharing and establishing a closed-loop management system—encompassing early warning of violations, rectification and tracking, and insurance rewards and punishments—the platform would effectively block multi-factor coupling risk chains at their source.

6. Conclusions

This paper proposes a marine traffic accident cause analysis framework that integrates a BERT + BiLSTM deep learning model with the Apriori association rule algorithm. By leveraging advanced NLP techniques and multi-factor association mining, the framework

efficiently classifies unstructured accident reports with an accuracy of 89.8% and systematically reveals complex causal relationships. This study identifies human factors (e.g., improper operation, fatigue), management deficiencies (e.g., owner malpractice), vessel conditions (e.g., equipment failure), and environmental factors (e.g., adverse sea conditions) as primary drivers of accidents. Moreover, high-confidence association rules (e.g., “fatigue -> improper use of equipment” and “illegal modification -> unseaworthy vessel”) demonstrate the significant impact of multi-factor interactions on accident severity. Spatial-temporal analysis further indicates that accident risks are higher at night, with collision and shipwreck incidents exhibiting seasonal trends, peaking in summer and winter, respectively. These findings provide a targeted basis for risk prevention and control measures in China’s coastal regions.

Despite these achievements, this study has several limitations. First, the data source is confined to accident reports publicly released by the China Maritime Safety Administration, which may not fully capture incident characteristics in other regions or unreported events. Additionally, the dataset’s concentration on maritime accidents limits the model’s classification performance for other accident domains. Furthermore, the available datasets restrict the model’s generalization capability, particularly in covering extremely rare events such as catastrophic accidents.

Future research should address these limitations by expanding the dataset to include time series data that capture the dynamic evolution of causality—such as real-time ship monitoring data (e.g., sensor logs) and crew behavior records—to construct a more comprehensive multi-modal database. Additionally, promoting the application and validation of unmanned ship technology during critical operational phases could further enhance maritime safety. By deepening data-driven approaches and fostering cross-disciplinary collaboration, future research is expected to improve the precision and foresight of maritime safety governance, thereby providing robust scientific support for the sustainable development of the global shipping industry.

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Notes

- ¹ [https://commercial.allianz.com/content/dam/onemarketing/commercial/commercial/reports/ Allianz Global Corporate & Specialty \(AGCS\) -Safety-Shipping-Review-2023.pdf](https://commercial.allianz.com/content/dam/onemarketing/commercial/commercial/reports/Allianz%20Global%20Corporate%20&%20Specialty%20(AGCS)%20Safety-Shipping-Review-2023.pdf).

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